



DSA MARATHON

C++ Fundamentals Cheat Sheet

One-page reference for syntax, operators, and common gotchas

1. Basic Program Structure

`#include <iostream>` → Enables cin, cout, cerr, clog.

`using namespace std;` → Avoids writing std::.

`int main()` → Entry point of every C++ program.

`return 0;` → Successful execution.

2. Input & Output

Function / Keyword	Purpose
<code>cin >></code>	Take input
<code>cout <<</code>	Print output
<code>endl</code>	New line + flush buffer
<code>\n</code>	New line (faster)
<code>getline(cin, str)</code>	Read complete line with spaces
<code>cin.ignore()</code>	Removes leftover newline before getline()

Keyboard → cin → Variables → Processing → cout → Screen

3. Variables

Declaration: `int x;`

Initialization: `int x = 10;`

Assignment / Update: `x = 20; x += 5;`

Common Types: int, long long, float, double, char, bool, string

4. Arithmetic Operators

+ (Add) – (Subtract) * (Multiply) / (Divide) % (Remainder)

`5/2 = 2` (Integer Division)

`5.0/2 = 2.5` (Floating Point Division)

5. Increment & Decrement

Pre Increment: `++x` → Increment first.

Post Increment: `x++` → Use then increment.

Pre Decrement: `--x`

Post Decrement: `x--`

6. Comments

`//` Single-line comment

`/* ... */` Multi-line comment

7. Types of Operators

Category	Operators
Arithmetic	+ - * / %
Relational	== != < > <= >=
Logical	&& !
Assignment	= += -= *= /= %=
Increment/Decrement	++ --
Bitwise	& ^ ~ << >>
Conditional	?: (Ternary)

8. Relational & Logical Operators

Relational: Compare values and return true(1) / false(0).

== Equal **!=** Not Equal **<** **>** **<=** **>=**

Logical:

&& → True only if both conditions are true.

|| → True if any one condition is true.

! → Reverses the boolean value.

9. char & ASCII

Character	ASCII
A	65
Z	90
a	97
z	122
0	48
9	57
Space	32

ASCII of character: `static_cast<int>(ch)`

Convert digit to integer: `digit - '0'`

10. Typecasting

Implicit: `double d = 10;`

Explicit: `static_cast<double>(a)`

Useful to avoid integer division.

11. Operator Precedence (High → Low)

`()` → `Unary(++ -- !)` → `*` `/` `%` → `+` `-` → `Relational(< > <= >=)` → `Equality(== !=)` → `&&` → `||` → `Assignment(=)`

12. Frequently Used Built-in Functions

`sizeof(x)` → Returns size in bytes.

`getline()` → Read full string.

`static_cast<type>()` → Safe type conversion.

13. Common Mistakes

WATCH OUT FOR

- Using `'='` instead of `'=='`.
- Forgetting `';`.
- Mixing `cin >>` and `getline()` without `cin.ignore()`.
- Confusing `x++` with `++x`.
- Forgetting integer division behavior.
- Ignoring operator precedence; use parentheses when needed.

Quick Revision

AT A GLANCE

- Program → `main()` | Input → `cin` | Output → `cout` | Full line → `getline()`
- New line → `\n` | Variables → Declaration + Initialization
- Operators → Arithmetic, Relational, Logical, Assignment, Bitwise
- `char` ↔ ASCII | `static_cast` for conversions | Parentheses have highest precedence.